

Mathematical Physics I (Fall 2025): Homework #1 Solution

Due Sep. 19, 2025 (Fri, 23:00pm)

[0.5 pt each, total 5 pts]

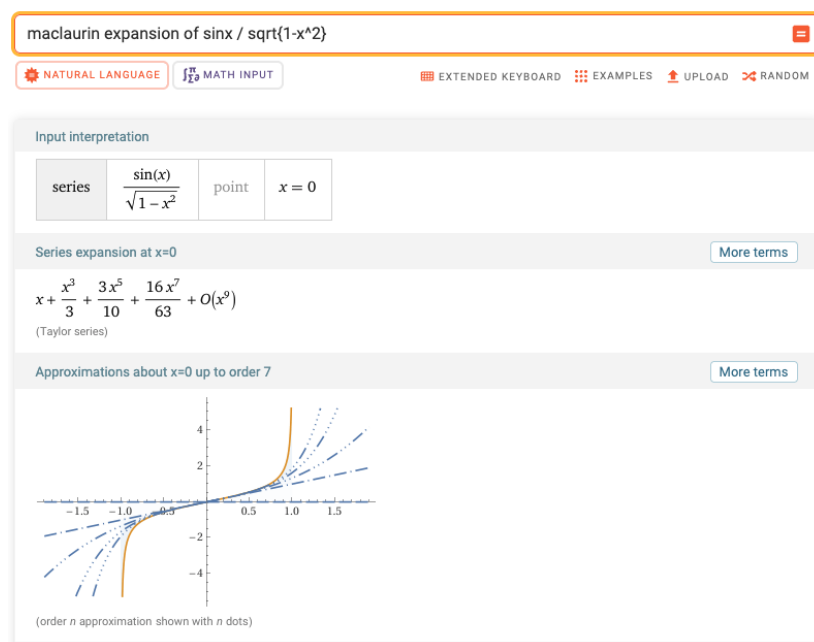
1. Boas Chapter 1, Problem 10.21

- The series converges for $0 \leq x \leq 1$ according to the ratio test.

2. Boas Chapter 1, Problem 13.36

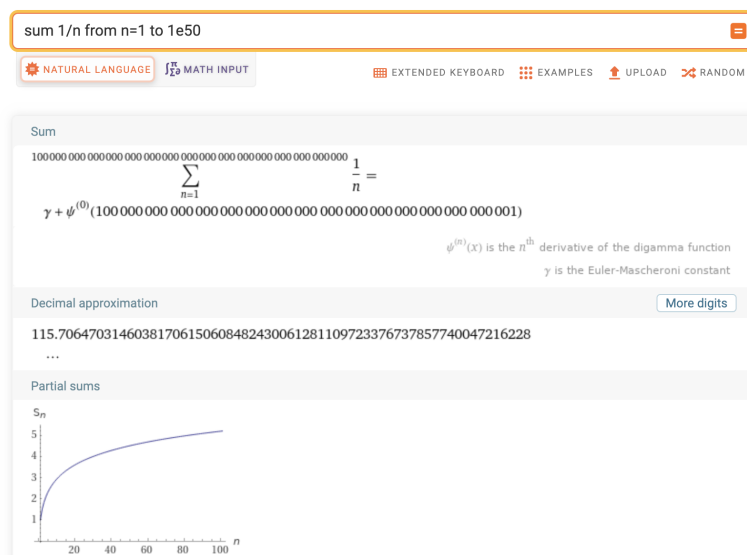
(Note: For Problem 13.36, read the instruction in the textbook carefully; among others, you have been asked to compare your result with a computer solution as well.)

- From Eq.(13.1) and (13.5), you get the binomial series expansions, $\sin x \simeq x - \frac{x^3}{3!} + \frac{x^5}{5!}$ and $(1 - x^2)^{\frac{1}{2}} \simeq 1 - (\frac{1}{2})x^2 + \frac{(\frac{1}{2})(-\frac{1}{2})}{2!}x^4$. Then, in a manner similar to Example 13.B2, the general terms of the Maclaurin series of the given integrand are found as $\frac{\sin x}{\sqrt{1-x^2}} \simeq x + \frac{1}{3}x^3 + \frac{3}{10}x^5$. Integrating both sides, you find $\int_0^u \frac{\sin x}{\sqrt{1-x^2}} dx \simeq \frac{1}{2}u^2 + \frac{1}{12}u^4 + \frac{1}{20}u^6$.



(Note: For Problem 16.1, it is helpful to draw a diagram that considers the balance between forces and torques, a so-called *extended free-body diagram*.)

- 
- WolframAlpha® computational intelligence™


$$\bullet \left\{ 16 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) \right\}^{\frac{1}{4}} = \left(16e^{\frac{2\pi}{3} + 2n\pi} \right)^{\frac{1}{4}} = 2e^{\frac{\pi}{6} + \frac{n\pi}{2}} = \{ \sqrt{3} + i, -1 + i\sqrt{3}, -\sqrt{3} - i, 1 - i\sqrt{3} \}.$$

(Note: For Problem 14.24, compare your result with a computer solution. Resolve or explain any disagreement in the two results.)

- (a-1) Using the definition of complex powers in Eq.(14.1), $[(-i)^{2+i}]^{2-i} = [e^{(2+i)\ln(-i)}]^{2-i} = [e^{(2+i)i(\frac{3\pi}{2}+2n\pi)}]^{2-i} = [e^{2i(\frac{3\pi}{2}+2n\pi)} \cdot e^{-(\frac{3\pi}{2}+2n\pi)}]^{2-i} = [-e^{-(\frac{3\pi}{2}+2n\pi)}]^{2-i} = e^{(2-i)\ln[-e^{-(\frac{3\pi}{2}+2n\pi)}]} = e^{(2-i)[i(\pi+2n\pi)+\ln[e^{-(\frac{3\pi}{2}+2n\pi)}]]} = e^{2\ln[e^{-(\frac{3\pi}{2}+2n\pi)}]+(\pi+2n\pi)} \cdot e^{i(-\ln[e^{-(\frac{3\pi}{2}+2n\pi)}]+2(\pi+2n\pi))}$.
- (a-2) Meanwhile, $(-i)^{(2+i)(2-i)} = -i$, which can be confirmed with the computer solution below. Note that, from our analytic calculations and computer solutions, your answer in (a-2) is a special case in your answer in (a-1).

$((-i)^{2+i})^{2-i}$

 NATURAL LANGUAGE  MATH INPUT  EXTENDED KEYBOARD  EXAMPLES  UPLOAD  RANDOM

Assuming i is the imaginary unit | Use i as a [variable](#) instead

Input
 $((-i)^{2+i})^{2-i}$
 i is the imaginary unit

Multivalued result [More](#) [Approximate forms](#) [Hide details](#)

General form
 $((-i)^{2+i})^{2-i} = e^{(1+2i)\arg((-i)^{2+i})+(2+4i)\pi n+(1-i/2)\pi}$ for $n \in \mathbb{Z}$
 (the choice of n is determined by the branch of the logarithm used for complex exponentiation)

Details
 $((-i)^{2+i})^{2-i} = e^{(2-i)\log((-i)^{2+i})}$
 $= \exp((2-i)(\operatorname{Log}((-i)^{2+i}) + 2i\pi n))$
 $= \exp((2-i)(i \tan^{-1}(\operatorname{Re}((-i)^{2+i}), \operatorname{Im}((-i)^{2+i})) + \frac{1}{2} \log(\operatorname{Im}((-i)^{2+i})^2 + \operatorname{Re}((-i)^{2+i})^2) + 2i\pi n))$
 $= e^{(1+2i)\arg((-i)^{2+i})+(2+4i)\pi n+(1-i/2)\pi}$
 (Log is the principal logarithm and n is determined by the branch chosen for log)

Values

n	-2	-1	0	1	2
value	$e^{(1+2i)\arg((-i)^{2+i})+(2+4i)\pi(-2)+(1-i/2)\pi}$	$e^{(1+2i)\arg((-i)^{2+i})+(2+4i)\pi(-1)+(1-i/2)\pi}$	$e^{(1+2i)\arg((-i)^{2+i})+(2+4i)\pi(0)+(1-i/2)\pi}$	$e^{(1+2i)\arg((-i)^{2+i})+(2+4i)\pi(1)+(1-i/2)\pi}$	$e^{(1+2i)\arg((-i)^{2+i})+(2+4i)\pi(2)+(1-i/2)\pi}$

$(-i)^{(2+i)(2-i)}$

 NATURAL LANGUAGE  MATH INPUT  EXTENDED KEYBOARD  EXAMPLES  UPLOAD  RANDOM

Assuming i is the imaginary unit | Use i as a [variable](#) instead

Input
 $(-i)^{(2+i)(2-i)}$
 i is the imaginary unit

Result [Step-by-step solution](#)

$-i$

Alternate complex forms

$\cos\left(-\frac{\pi}{2}\right) + i \sin\left(-\frac{\pi}{2}\right)$ [Radians](#) [Approximate forms](#)

6. Boas Chapter 2, Problem 15.6

(Note: For Problem 15.6, read the instruction in the textbook carefully; that is, you have been asked to compare your result with a computer solution as well. Resolve or explain any disagreement in results.)

- With $u \equiv e^z$, you get $\tanh z = -i = \frac{e^z - e^{-z}}{e^z + e^{-z}} = \frac{u - u^{-1}}{u + u^{-1}} \rightarrow u^2 = \frac{1-i}{1+i} = -i \rightarrow z = \ln u = \ln[\pm(-i)^{\frac{1}{2}}] = \ln[e^{i(\frac{3}{2}\pi + 2n\pi) \cdot \frac{1}{2}}] = i(n\pi + \frac{3}{4}\pi).$

inversehyperbolictan(-i)

 NATURAL LANGUAGE  MATH INPUT  EXTENDED KEYBOARD  EXAMPLES  UPLOAD  RANDOM

Assuming i is the imaginary unit | Use i as a [variable](#) instead

Input

$\tanh^{-1}(-i)$

$\tanh^{-1}(x)$ is the inverse hyperbolic tangent function
i is the imaginary unit

Exact result

$-\frac{i\pi}{4}$

Decimal approximation [More digits](#)

-0.78539816339744830961566084581987572104929234984377645524373614... i

7. Boas Chapter 2, Problem 17.21

(Note: For Problem 17.21, verify only the first equality, $\cosh^{-1}z = \ln(z \pm \sqrt{z^2 - 1})$.)

- $\cosh^{-1}z = \omega \rightarrow$ With $u \equiv e^\omega$, you find $z = \cosh \omega = \frac{e^\omega + e^{-\omega}}{2} = \frac{u + u^{-1}}{2} \rightarrow u^2 - 2zu + 1 = 0 \rightarrow \omega = \ln u = \ln(z \pm \sqrt{z^2 - 1}) = \cosh^{-1}z$.

8. Boas Chapter 3, Problem 2.8

(Note: For Problem 2.8, again, read the instruction in the textbook carefully.)

- The rank of the augmented matrix is 2 while there are 3 unknowns.

row reduce {{-1,1,-1,4},{1,-1,2,3},{2,-2,4,6}}

 NATURAL LANGUAGE  MATH INPUT  EXTENDED KEYBOARD  EXAMPLES  UPLOAD  RANDOM

Input interpretation

row reduce $\begin{pmatrix} -1 & 1 & -1 & 4 \\ 1 & -1 & 2 & 3 \\ 2 & -2 & 4 & 6 \end{pmatrix}$

Result [Step-by-step solution](#)

$\begin{pmatrix} 1 & -1 & 0 & -11 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

9. Consider a state of a system (e.g., a particle in a box, or a quantized oscillator) whose energy when occupied by a single particle is ϵ_0 . In the quantum statistical distribution of bosons with unspecified total number of particles (so-called *Bose-Einstein* statistics), the expected total energy of all the particles in this state is found to be

$$\bar{\epsilon} = \frac{\sum_{n=1}^{\infty} n\epsilon_0 e^{-\frac{n\epsilon_0}{k_B T}}}{\sum_{n=0}^{\infty} e^{-\frac{n\epsilon_0}{k_B T}}}, \quad (1)$$

where k_B is the Boltzmann constant, and T is the temperature. (a) After briefly discussing how this formula is acquired, show that the ratio becomes

$$\bar{\epsilon} = \frac{\epsilon_0}{e^{\frac{\epsilon_0}{k_B T}} - 1}$$

by first identifying Eq.(1)'s denominator as a binomial expansion of $\frac{1}{1-x}$ with $x = e^{-\frac{\epsilon_0}{k_B T}}$, and its numerator as a constant times $\frac{x}{(1-x)^2}$. (b) Using a power series expansion, show that $\bar{\epsilon}$ reduces to $k_B T$ in the classical limit of $k_B T \gg \epsilon_0$.

(Note: You may want to briefly review the classic textbooks in statistical mechanics such as Schroeder or Reif. To receive full credit, you are asked to come up with a short paragraph about what Eq.(1) means, without having to laboriously derive the equation in great detail. If needed, you must reference your sources appropriately with a proper citation convention, but your answer must still be your own work in your own words. To access the electronic resources — e.g., academic journals — off-campus via SNU library's proxy service, see <http://library.snu.ac.kr/using/proxy>.)

- (a) By differentiating $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, you find $\frac{x}{(1-x)^2} = \sum_{n=1}^{\infty} n x^n$, which leads to

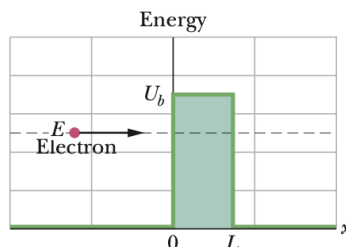
$$\bar{\epsilon} = \frac{\sum_{n=1}^{\infty} \epsilon_0 n x^n}{\sum_{n=0}^{\infty} x^n} = \frac{\epsilon_0 x}{1-x} = \frac{\epsilon_0}{x^{-1} - 1}.$$

- (b) With $y = \frac{\epsilon_0}{k_B T} \ll 1$, you get $\bar{\epsilon} = \frac{\epsilon_0}{e^y - 1} \simeq \frac{\epsilon_0}{1+y-1} = k_B T$.

10. In quantum mechanics, consider the transmission coefficient T of an electron matter wave (of mass m and energy E) incident on a rectangular potential barrier of height U_b (see figure). Briefly explain why the following expression must be evaluated to determine the probability of the electron being transmitted through the barrier and detected on the other side:

$$T^{-1} = \left| \frac{(k + i\kappa)^2 - (k - i\kappa)^2 e^{-2\kappa L}}{4ik\kappa e^{-\kappa L} e^{-ikL}} \right|^2 = 1 + \left(\frac{k^2 + \kappa^2}{2k\kappa} \right)^2 \sinh^2(\kappa L), \quad (2)$$

where $k \equiv \frac{\sqrt{2mE}}{\hbar}$ and $\kappa \equiv \frac{\sqrt{2m(U_b - E)}}{\hbar}$ are real numbers. Prove the second equality using complex algebra.



[Adapted from Halliday & Resnick, Chapter 38.9]

(Note: You may want to review the freshman physics textbooks such as Halliday & Resnick (Chapters 38.8 and 38.9), or the quantum mechanics textbooks such as Griffiths & Schroeter

(Chapter 2.6 and problems therein) or Landau & Lifshitz. To receive full credit, you should include a brief explanation — without a detailed derivation — of how the absolute square of a complex number like Eq.(2) arises in calculating the transmission probability.)

- The wave function $\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & \text{for } x < 0, \\ Ce^{-\kappa x} + De^{\kappa x}, & \text{for } 0 < x < L, \\ Fe^{ikx}, & \text{for } x > L \end{cases}$ describes an electron in the

three regions in the figure. Using the four boundary conditions – continuity of ψ and $\frac{d\psi}{dx}$ at $x = 0$ and L – on five unknowns, one can get the ratios $\frac{D}{C}$, $\frac{B}{A}$, and $\frac{F}{A}$. In particular, one finds the inverse of the transmission coefficient as

$$T^{-1} = \left| \frac{A}{F} \right|^2 = \left| \frac{(k + i\kappa)^2 - (k - i\kappa)^2 e^{-2\kappa L}}{4ik\kappa e^{-\kappa L} e^{-ikL}} \right|^2.$$

- Now with complex algebra,

$$\begin{aligned} T^{-1} &= \left| \frac{(k + i\kappa)^2 e^{\kappa L} - (k - i\kappa)^2 e^{-\kappa L}}{4ik\kappa e^{-ikL}} \right|^2 = \left| \frac{(k^2 - \kappa^2 + 2ik\kappa)e^{\kappa L} - (k^2 - \kappa^2 - 2ik\kappa)e^{-\kappa L}}{4ik\kappa e^{-ikL}} \right|^2 \\ &= \left| \frac{2(k^2 - \kappa^2) \sinh(\kappa L) + 4ik\kappa \cosh(\kappa L)}{4ik\kappa e^{-ikL}} \right|^2 = \frac{(k^2 - \kappa^2)^2 \sinh^2(\kappa L) + 4k^2 \kappa^2 \cosh^2(\kappa L)}{4k^2 \kappa^2} \\ &= \frac{(k^2 + \kappa^2)^2 \sinh^2(\kappa L) + 4k^2 \kappa^2 \{\cosh^2(\kappa L) - \sinh^2(\kappa L)\}}{4k^2 \kappa^2} \\ &= 1 + \left(\frac{k^2 + \kappa^2}{2k\kappa} \right)^2 \sinh^2(\kappa L). \end{aligned}$$